Solving Diffusion Equations with Neumann Boundary Conditions

1.] Write down the linear ODE system that corresponds to this IVBP:

$$\begin{cases} u_t = k u_{xx} & \text{for } x \in (0, L),\ t > 0 \\ u_x(0,t) = 0 & \text{for } x = 0,\ t > 0 \\ u_x(L,t) = 0 & \text{for } x = L,\ t > 0 \\ u(x,0) = f(x) & \text{for } x \in (0, L), t = 0 \end{cases}$$

Implement the linear system into Matlab and solve.

2.] Write down the linear ODE system that corresponds to this IVBP:

$$\begin{cases} u_t = k u_{xx} - \alpha u & \text{for } x \in (0, L),\ t > 0 \\ u_x(0,t) = 0 & \text{for } x = 0,\ t > 0 \\ u_x(L,t) = 0 & \text{for } x = L,\ t > 0 \\ u(x,0) = f(x) & \text{for } x \in (0, L), t = 0 \end{cases}$$

Implement the linear system into Matlab and solve.

3.] Write down the nonlinear ODE system that corresponds to this IVBP:

$$\begin{cases} u_t = ku_{xx} + \alpha u\left(1 - \dfrac{u}{m}\right) & \text{for } x \in (0, L),\ t > 0 \\ u_x(0, t) = 0 & \text{for } x = 0,\ t > 0 \\ u_x(L, t) = 0 & \text{for } x = L,\ t > 0 \\ u(x, 0) = f(x) & \text{for } x \in (0, L), t = 0 \end{cases}$$

Implement the linear system into Matlab and solve.